Why the emission methods disagree

137 episodes | 118 sensor-date profiles | 4,800 five-minute records.
Readability revision: 14 October 2026.

The CO_2/NO_x gap persists after controlling for predicted travel time. The two-speed rate allocation remains the strongest source of disagreement with the detector-based reference. Delay calibration matters more for CO and HC.

Common-sample emission errors

Table 1. WAPE (%) over the same 137 episodes.

Pollutant	S	S_p	R	M	A
CO_2	5.94	9.72	10.13	22.38	20.32
NO_x	12.43	14.84	13.05	47.73	42.71
CO	6.12	9.91	8.27	11.19	9.82
HC	4.24	9.57	9.52	12.61	11.65

S: observed mean speed; S_p: predicted mean speed; R: reconstructed trip speeds; M: two-speed lookup rates; A: two-speed cubic rates. O is the recorded-speed reference.

S uses the observed mean speed and exactly inherits the reference's total vehicle-hours. S_p is an additional diagnostic using the same *predicted* vehicle-hours as M. This separates part of the traffic-information advantage of S from the rate representation.

Table 2. Episodes with larger absolute error than the stated comparator.

Pollutant	M worse than S	A worse than S	M worse than S_p
CO_2	121/137	121/137	99/137
NO_x	122/137	121/137	108/137
CO	64/137	70/137	52/137
HC	94/137	89/137	72/137

"Worse" means $|E_{m,j} - E_{O,j}| > |E_{c,j} - E_{O,j}|$ for episode j and the same pollutant, where c is the comparator. Rankings must be labeled separately for each pollutant.

Sample and interpretation

The sample contains 20 Arizona episodes from 18 monthly profiles and 117 California episodes from 100 released profiles. All 137 are episodes from fitted average profiles, not held-out daily observations. The audit preserves the original admission mask and all reported emission totals; it refits no parameter and excludes no additional episode.

The benchmark treats five-minute point speeds as whole-link speeds. These results explain agreement with that benchmark; they do not establish which method is closest to actual vehicle trajectories or measured exhaust.

Metric definitions and notation

WAPE is not the error between grand totals

For one pollutant, let $E_{m,j}$ be method m's total emission for episode j, and let $E_{O,j} > 0$ be its reference emission. With $N = 137$,

$$\mathrm{WAPE}_m = 100\,\frac{\sum_{j=1}^{N} |E_{m,j} - E_{O,j}|}{\sum_{j=1}^{N} E_{O,j}},$$

$$B_m = 100\,\frac{\sum_{j=1}^{N} (E_{m,j} - E_{O,j})}{\sum_{j=1}^{N} E_{O,j}}.$$

WAPE adds *absolute episode errors*. The signed total bias B_m permits overestimates and underestimates to cancel. Larger-reference episodes have more weight in WAPE.

Notation used consistently throughout this report

Symbol	Meaning and units
$i,\ j,\ m$	Five-minute bin, episode, and emission method.
$d_i,\ D = \sum_i d_i$	Bin count and episode count (vehicles on the stated lane basis).
$L,\ C$	Link length (mi) and hourly capacity (vehicles/h, same lane basis as D).
$v_i,\ v_f,\ \widehat{v}_q$	Recorded bin speed, free-flow speed, and fitted queue speed (mph).
$T_f = L/v_f$	Free-flow link travel time per vehicle (h).
$P,\ \widehat{P}$	Observed and fitted congestion-episode durations (h).
$w_{\max},\ \widehat{w}_{\max}$	Observed and fitted peak link delays per vehicle (h).
$\bar{w},\ \widehat{\bar{w}}$	Observed and fitted count-weighted mean link delays (h).
$\theta_{\mathrm{obs}},\ \theta$	Observed and fitted mean-to-peak delay ratios (dimensionless).
f_d, n, f_p, s	Fitted coefficients and exponents; f_d has units h^{1-n}, f_p has units h^{-s}; n, s are dimensionless.
$H_O,\ \widehat{H}$	Observed and fitted total vehicle-hours (vehicle h).
$\bar{v},\ \widehat{\bar{v}}$	Equivalent observed and fitted mean trip speeds (mph).
$r_0(v),\ r_c(v)$	Lookup and cubic emission rates (g/(vehicle h)).
$r_m^{\mathrm{eff}} = E_m/H_m$	Method-specific effective emission rate (g/(vehicle h)).
$E_m,\ \Gamma$	Episode emission (g), and the two-speed delay multiplier (g/(vehicle h)).

An episode subscript j and pollutant label are suppressed in the within-episode equations. Hats denote fitted traffic quantities. S_{p} corresponds to `S_pred` in the CSV files.

Delay prediction and its ingredients

Observed and fitted quantities

The reference uses the same link length and free-flow baseline as the model:

$$D = \sum_i d_i, \qquad H_O = \sum_i d_i \frac{L}{v_i}, \qquad \bar{w} = \frac{H_O}{D} - T_f,$$

$$w_{\max} = \frac{L}{\min_i v_i} - T_f, \qquad \widehat{P} = \max\{f_d (D/C)^n, D/C\}\,.$$

$$\widehat{w}_{\max} = T_f f_p \widehat{P}^{\,s}, \qquad \widehat{\bar{w}} = \theta \widehat{w}_{\max}, \qquad \widehat{H} = D(T_f + \widehat{\bar{w}}).$$

Table 3. Traffic errors in the retained 137 episodes.

Traffic quantity	WAPE (%)	Interpretation
Episode duration	6.17	Estimated duration is comparatively accurate.
Peak delay	33.83	Both targets use the same free-flow baseline.
Peak delay with observed duration	32.31	Replacing predicted duration changes little.
Mean delay per episode	28.61	Scores the per-vehicle mean-delay targets.
Total vehicle delay	28.24	Multiplies mean delay by the episode count.
Total vehicle-hours	14.77	Includes free-flow time; total bias is −1.85%.

Separate duration, severity, and mean-to-peak effects

Define $w_{\max}^{\text{cond}} = T_f f_p P^s$, using observed duration but the fitted severity coefficients. The following identity holds episode by episode:

$$\begin{aligned} \widehat{\bar{w}} - \bar{w} = {} & (\theta - \theta_{\text{obs}}) w_{\max} \\ & + \theta\big(w_{\max}^{\text{cond}} - w_{\max}\big) \\ & + \theta\big(\widehat{w}_{\max} - w_{\max}^{\text{cond}}\big). \end{aligned}$$

Multiplying each term by D and summing gives **+157.59, −251.74, and +10.11 vehicle-hours**, respectively; the net error is −84.04 vehicle-hours. The comparatively small duration term agrees with the conditional peak-delay test in Table 3.

These are ordered signed contributions. Their absolute values do not add to a causal percentage of prediction error.

Equal travel time, different emission rates

Define the equivalent fitted mean speed as $\widehat{\bar{v}} = DL/\widehat{H}$. The fraction of fitted trip time spent at free-flow speed is

$$\lambda_f = \frac{\widehat{\bar{v}} - \widehat{v}_q}{v_f - \widehat{v}_q}.$$

The two-speed method combines the operating rates using that time share:

$$r_M^{\text{eff}} = \lambda_f r_0(v_f) + (1 - \lambda_f) r_0(\widehat{v}_q),$$

$$E_M - E_{S_{\text{p}}} = \widehat{H}\left[r_M^{\text{eff}} - r_0(\widehat{\bar{v}})\right].$$

Geometrically, M uses the straight chord between the two operating-speed rates. S_{p} uses the lookup curve at the equivalent mean speed. Count, distance, and predicted vehicle-hours are identical.

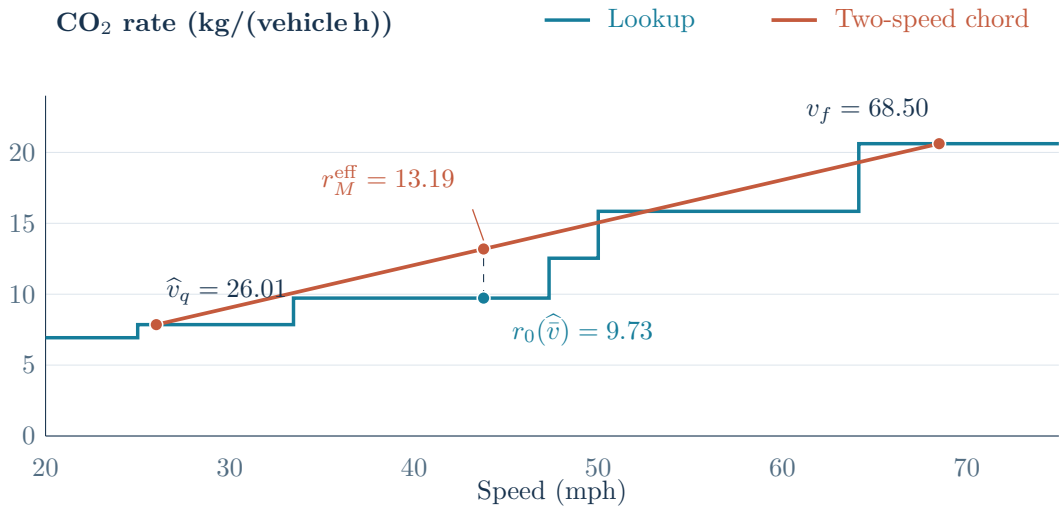


Figure 1. The same-time rate difference for D12 I-5 north, L5N-028, AM 07:35–09:15. The vertical dashed segment is the effective-rate difference at the fitted mean speed. Rates are displayed in kg/(vehicle h).

A case with accurate delay but large two-speed error

Observed mean speed is 44.56 mph, fitted mean speed is 43.78 mph, and recorded speed CV is 0.0418. Mean-delay error is only +5.12%. Nevertheless, CO_2 absolute percentage error is 2.59% for S and 34.47% for M. The chord rate is 35.62% above the lookup rate at the fitted mean speed.

Across all 137 episodes, the median free-flow time share is 34.8%, while the median free-flow emission share is 60.0% for CO_2 and 76.3% for NO_x. This identifies a representation difference; it does not imply that physical free-flow travel should be removed.

How emission errors accumulate

Exact signed accounting

For each episode and pollutant, the successive differences telescope:

$$E_A - E_O = (E_S - E_O) + (E_{S_\mathrm{p}} - E_S) + (E_R - E_{S_\mathrm{p}}) + (E_M - E_R) + (E_A - E_M).$$

Table 4. Sum of each signed component divided by total reference emissions.

Signed contribution (pp)	CO_2	NO_x	CO	HC
Observed mean-rate aggregation: $E_S - E_O$	-4.60	-9.56	-2.86	-2.51
Fitted mean time and speed: $E_{S_\mathrm{p}} - E_S$	+1.03	+4.48	-1.22	-0.45
Reconstructed speed variation: $E_R - E_{S_\mathrm{p}}$	+7.78	+15.99	+2.28	+2.87
Two-speed allocation: $E_M - E_R$	+16.53	+35.63	+3.90	+5.08
Cubic rate approximation: $E_A - E_M$	-1.99	-4.39	-0.97	-0.82
Sum: total bias of A	+18.76	+42.15	+1.13	+4.18

Entries are percentage points (pp). They sum to total bias, not WAPE. Cancellation occurs across steps and episodes. R and M share fitted vehicle-hours up to integration precision.

Separate vehicle-hours from effective rate

Using $r_O^{\mathrm{eff}} = E_O/H_O$ and $r_M^{\mathrm{eff}} = E_M/\widehat{H}$ gives another exact identity:

$$E_M - E_O = (\widehat{H} - H_O)\frac{r_M^{\mathrm{eff}} + r_O^{\mathrm{eff}}}{2} + (r_M^{\mathrm{eff}} - r_O^{\mathrm{eff}})\frac{\widehat{H} + H_O}{2}.$$

Table 5. Signed product components, normalized by total reference emissions.

Signed contribution (pp)	CO_2	NO_x	CO	HC
Vehicle-hours component	-1.40	-0.90	-1.21	-1.39
Effective-rate component	+22.15	+47.44	+3.31	+6.39

The effective-rate component includes operating-speed representation, rate nonlinearity, and calibration-dependent queue speed. It is not a pure emission-function error. Both decompositions are mathematical accounting devices, not independently identified causal effects.

Does correcting mean delay help?

A controlled diagnostic

Hold the fitted queue speed $\widehat{v}_q$ and both lookup rates fixed. The two-speed rate multiplier is

$$\Gamma = \frac{v_f r_0(\widehat{v}_q) - \widehat{v}_q\, r_0(v_f)}{v_f - \widehat{v}_q}.$$

Replace only the fitted mean delay with the observed mean delay:

$$E_M^{\text{corr}} = D\left[T_f r_0(v_f) + \bar{w}\,\Gamma\right],$$
$$E_M - E_M^{\text{corr}} = D(\widehat{\bar{w}} - \bar{w})\Gamma.$$

Table 6. Original and corrected errors on the same 133 feasible episodes.

Pollutant	Original M WAPE (%)	Corrected M WAPE (%)
CO_2	22.53	21.82
NO_x	48.27	46.78
CO	10.85	7.50
HC	12.29	7.61

Correcting mean delay changes the CO_2/NO_x errors very little, while reducing CO and HC errors substantially. Mean-delay calibration alone therefore cannot explain the CO_2/NO_x discrepancy.

Which corrections are physically feasible?

Nonnegative *mean* free-flow and queue times require

$$0 \le \bar{w} \le L\left(\frac{1}{\widehat{v}_q} - \frac{1}{v_f}\right).$$

This condition holds for 133 episodes. For the other four, the observed mean speed is below the fitted queue speed, so the correction would require negative free-flow time.

A stricter test retains the fitted duration and shape, rescales peak delay to $\bar{w}/\theta$, and checks the entire profile. Only 110 episodes then satisfy the finite-link peak constraint; the corrected arrival rates remain nonnegative in all 137.

Independent check using already accurate delay predictions

Among the 34 episodes with mean-delay absolute percentage error at most 10%, M remains worse than S in 31 CO_2 and 31 NO_x cases. CO_2 WAPE is 20.09% versus 5.96%; NO_x WAPE is 43.39% versus 12.02%.

The corrected values are counterfactual diagnostics, not newly calibrated forecasts. Table 6 uses the feasible 133-case sample on both sides; it must not be compared as though it used all 137 cases.

Profile tests and episode-level files

Are flat or short episodes the explanation?

Table 7. CO_2 results by recorded-profile characteristics.

Condition	N	M worse than S	S / M WAPE (%)
Flat recorded speeds: CV ≤ 0.10	81	73/81	4.73 / 23.91
More variable speeds: CV > 0.10	56	48/56	6.45 / 21.74
Short episodes: $P \leq 1$ h	26	23/26	8.06 / 20.48
Longer episodes: $P > 1$ h	111	98/111	5.83 / 22.48

Flatness here means little temporal variation in the recorded five-minute speeds, using count-weighted CV. It does not establish flat individual trajectories. In the same 81 flat episodes, CO WAPE is 6.51% for M versus 6.59% for S: the finding is pollutant-specific.

No retained episode meets the illustrative shallow threshold $\bar{w}/T_f \leq 0.25$; the minimum is 0.370. Yet 79 episodes have mean delay at most 10 seconds because many links are short. Small absolute delay and weak relative congestion are distinct conditions.

The excess CO_2 absolute percentage error of M over S has weak rank associations with absolute trip-time percentage error (-0.045), speed CV (-0.086), and relative congestion severity (-0.038); the association with fitted queue speed is -0.389. These are descriptive and may reflect corridor differences, not causal effects.

Files for checking individual episodes

File	Contents
`episode_review_137.csv`	Short entry point: sorted CO_2 error gap, pollutant-specific comparison flags, and traffic diagnostics.
`episode_calculations_137.csv`	All 137 episodes, every method total, rates, time shares, errors, and admissibility flags.
`episode_pollutant_calculations_548.csv`	Narrower format: one episode-pollutant pair per row.
`column_definitions.csv`	Calculation definitions and units for the full episode table.
`comparison_metrics.csv`; `counterfactual_metrics.csv`	Pooled scores for explicitly named, matched samples.

Audit evidence. All saved O/S/R/M/A totals and observed traffic targets were reproduced from the archived inputs; maximum relative discrepancy was 6.44×10^{-13}. An independent calculation verified the comparisons and identities. No numerical result changed during this formatting revision.

Scope. These 137 admitted cases are a selected sample of 325 detected episodes. The analysis explains agreement with a point-speed reference; it does not validate physical vehicle trajectories or measured exhaust. Source hashes and detailed checks accompany the analysis package.